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A conceptual framework for supply chain digital twins – development and evaluation

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ABSTRACT
Digital twins, due to digitalisation, are increasingly adopted across industries. The supply chain industry, with its process-driven nature, collaboration of multiple stakeholders and dependability on real-time events, demands unique characteristics for the implementation of digital twins. While these twins can enhance supply chain transparency and responsiveness, there is a lack of an application framework. The aim of this paper is to address this absence of a specialised application framework. The framework outlines multiple layers, dimensions, and stakeholder-technology dependencies for early-phase adoption. It is designed to guide planners and operators of supply chains in planning, implementing, and introducing new supply chain digital twins. Using the design science approach as a research methodology, the framework is informed by literature and expert interviews. Its validity is confirmed through further expert evaluations, and the study concludes with key insights and guidelines derived from the framework.

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Digital twin; supply chain; framework; logistics; industry 4.0; design science

Introduction

Delays and disruptions in the supply chain pose major challenges for companies, particularly when operating globally (Ivanov and Dolgui 2020; Mahajan and Tomar 2021; Scheibe and Blackhurst 2018). They predominantly arise from limited visibility; for instance, third- or fourth-tier suppliers are often unknown, the exact location and status of shipments are unclear, or the scheduling order on terminals is not being communicated (Ageron, Bentahar, and Gunasekaran 2020; M. Barratt and Oke 2007). Further demand spikes due to increased customer orders and resource issues due to production interruptions or harvest shortfalls can place high flexibility requirements on the supply chain (Garcia and You 2015; Murray, Agard, and Barajas 2015). This potentially results in over- or understocking with inventory levels that are too high or too low. Additionally, shipping companies require alternative transport routes in case of major disruptions of global supply chains. One example of supply chain disruptions was the Suez Canal blockade in March 2021. The six-day blockade made it impossible for other ships to pass through the canal. Ships piled up in front of both canal entrances; on the day of unblocking, they numbered 370. In particular, companies that were able to react quickly and take the longer route around Africa were not stuck for several days (Ramos et al. 2021). Different types of unexpected events and disruptions can lead to different critical situations and pose challenges in the supply chain. In addition to these irregularities, requirements make it necessary that companies have recently started documenting their CO_2 emissions to prove their carbon footprint to their customers. Currently, the consequential

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environmental and social damages of resource procurement and production are often unknown (Gualandris et al. 2021; Hofmann, Schleper, and Blome 2018; Skene 2022).

To address the abovementioned challenges, various information systems exist comprising of diverse technologies, processes and support for human decision-making (Daneshvar Kakhki and Gargeya 2019). For instance, they include systems for tracking and tracing the movement of goods, for managing inventory levels such as supply chain management systems, or for forecasting expected times of arrival (ETA). However, no integrated system combines these technologies for end-to-end supply chain visibility and proactive management. A promising approach as a blueprint concept for such an integrated system is digital twins. Digital twins that have emerged in the course of digitalisation could potentially play an important role in addressing major supply chain challenges (Grabis, Stirna, and Zdravkovic 2020; Ivanov and Dolgui 2020). A digital twin is the virtual representation of a physical object in an information system. It enables benefits in monitoring and simulation of its counterpart and provides a basis for disruption management and real-time predictions (Dietz and Pernul 2020).

To date, digital twins have been mainly used for product life cycle. They are used during product design, runtime operation, maintenance and control, i.e. for electric vehicle development (Lo, Chen, and Zhong 2021). The use of digital twins in the supply chain is a new approach that aims to bring the benefits of digital twins to the supply chain. Among other applications, processes in production can be mapped.

While frameworks for the realisation of digital twins exist for different application areas such as product development (Tao et al. 2019; Yi et al. 2021) or manufacturing (Kunath and Winkler 2018; Onaji et al. 2022; Zheng, Yang, and Cheng 2019; Zhuang, Liu, and Xiong 2018), a framework for the supply chain area does not yet exist. This paper aims to fill this gap. We propose a framework for supply chain digital twins that is developed based on the design science methodology. It supports planners and operators of supply chains in planning, implementing, and introducing new supply chain digital twins. It is unique to supply chains because it considers its various dimensions, multi-tier organisational structure, and different actors. It ensures that all partial activities, such as transport or warehousing, are covered. Furthermore, this research addresses issues previously identified in existing literature, such as insufficient data integration (Petrova-Antonova and Ilieva 2019), unclear ownership of data (Andersson and Eckerwall 2019), lack of real-time control (Uhlemann, Schock et al. 2017), and cost and time issues (West and Blackburn 2017).

Overall, we address the following research questions: What are the essential components of supply chain digital twins? How can the components be structured, and how do they relate to each other? To answer these research questions, we draw on expert interviews for requirements development and evaluation. Our work makes three contributions. First, we contribute to the still early research of the adoption of the digital twin into the supply chain context. Second, the developed framework documents the different dimensions that are relevant in this context. Third, it proposes a new conceptual framework that can be applied to the development of various digital twins in the supply chain domain.

The article is structured as follows. First, in section two, the basics of supply chains as well as digital twins and relevant research are presented. Then, section three describes the methodology of our research. This is followed by the analysis, which is divided into a literature analysis of the different use cases and a qualitative requirements analysis based on expert interviews. The fifth section contains the developed framework. In the following sixth section, the evaluation of our artifact is explained. In the seventh section, the implications and guidelines are discussed. Finally, the article concludes with a summary and a look into future research opportunities.

Background

First, we look at the definitions of the supply chain and digital twin individually. Then, we extract the state-of-the-art supply chain digital twins from related surveys.

Supply chain

A supply chain is a complex and dynamic system consisting of suppliers and customers. It constitutes flows of goods and information to satisfy customer needs through collaboration (Beamon 1998; Lummus and Vokurka 1997). Supply chain management entails the strategic coordination of traditional business functions and tactical decisions across these business functions. The goal is to improve the long-term performance of individual companies and the supply chain as a whole. In the simplest case, the supply chain of a company only takes direct suppliers and direct customers into consideration. With the increasing maturity of supply chain management, the coverage is expanded up from raw material sourcing to different tiers of production and distribution up to the final consumer (Janvier-James 2011; Mentzer et al. 2001). Supply chain processes are described in the supply chain reference model (SCOR) (Huan, Sheoran, and Wan 2004). On a general level, it describes business processes that are performed within a company and between the players in a cross-company value chain. It includes the involved key processes of plan, source, make, deliver, (return), and flows of goods, information and payments. Various disruptions can occur in the supply chain that require decisions. To make these decisions quickly and effectively, it is necessary to have visibility across the supply chain and its interdependencies between the individual processes and actors. Decisions must be made as quickly as possible, potentially in real-time, to minimise the impact of disruptions. Real-time data with automated decision-making requires technologies and concepts such as the digital twin (Ivanov and Dolgui 2020).

Digital twin

The idea of digital twins has been around since the 1990s (Gelernter 1991), but only the Internet of Things concept has promoted its implementation (Tao and Zhang 2017). In the historical context, the concept of digital twins was coined first in the aviation industry and was used to develop and monitor parts of turbines via a software image (Grieves 2019). The first application was in product lifecycle management, where the digital twin supported the product from its design phase until its usage phase (Grieves 2019). The concept of digital twins was then extended for processes, such as the manufacturing process (Qi et al. 2018). A twin is an identical manifestation of an object. Digital indicates that its twin is not an object of the real world but data saved in an information system. Glaessgen and Stargel (2012) give the following definition: A digital twin is an integrated multiphysics, multiscale, probabilistic simulation of a product or system that uses the best available physical models, sensor updates, and the like, to mirror the life of its corresponding twin. In contrast to the definition of digital twins in manufacturing and lifecycle management, definitions specific to supply chain digital twins do exist. Ivanov et al. (2019) define a supply chain digital twin as a model that can represent the network state for any given moment in time and allow for complete end-to-end supply chain visibility to improve resilience and test contingency plans. The supply chain of digital twins covers the complete supply chain, from supplier to production to customer, including all intermediaries. The supply chain digital twin tracks all processes according to the SCOR model, as illustrated in Figure 1.

Various technologies support the digital twin. Different sensors, such as RFID or cameras, exist to collect data. For mapping the data, there are technologies such as XML, Automation ML or OPC UA. The data are processed using edge computing or blockchain technologies. The data are then transmitted via Bluetooth, WLAN, 5G, or Fieldbus (Fuller et al. 2020; Tao and Zhang 2017; M. Liu et al. 2021; Qi et al. 2021). With the help of these data, various applications, such as monitoring, simulation, and prediction, can be carried out (Z. Liu, Meyendorf, and Mrad 2018; M. Liu et al. 2021; Qi et al. 2021).

Today, digital twins are used in different application fields, from healthcare (Feng, Chen, and Zhao 2018) to the construction industry (Kaewunruen, Rungskunroch, and Welsh 2018; Qi et al. 2021). The area of the supply chain for digital twins often focuses on the production part, such

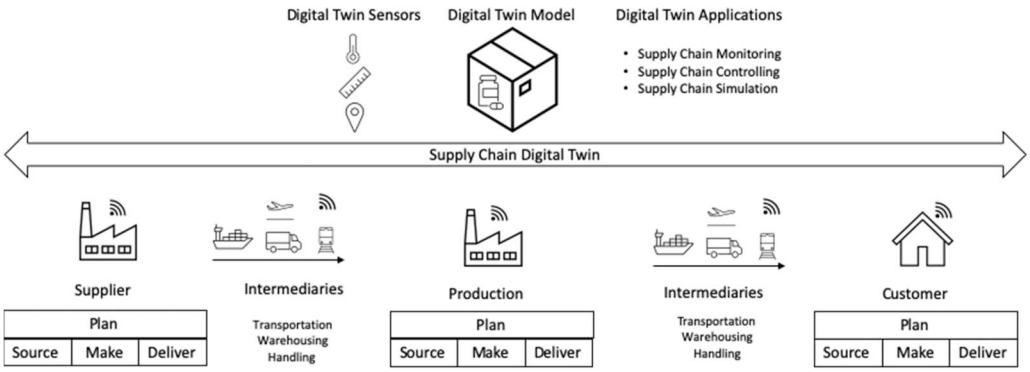

Figure 1. SCOR model and supply chain digital twin.

as shop-floor digital twins (Qi et al. 2018). However, supply chain digital twins should focus on the entire supply chain. They provide real-time data on products and processes in the supply chain and increase transparency (Srai et al. 2019).

In the domain of supply chain management, the integration of digital twin technologies necessitates a critical examination of data-centric challenges. Organisational challenges encompass issues like data access, where ensuring data availability across various stakeholders while adhering to privacy standards is complex (Birkel and Müller 2021). Data ownership presents legal and ethical considerations in terms of rights and usage, while data interpretation requires standardised methodologies to ensure consistency across different supply chain actors (Richey et al. 2016). Technically, challenges start with data collection from diverse sources, each using different technologies (Li et al. 2023; Vivi et al. 2019). Ensuring interoperability and secure data sharing is crucial, as is managing the storage of large data volumes, requiring scalable and secure solutions (Chen and Huang 2021; Lee et al. 2021). Data analysis demands sophisticated algorithms to extract actionable insights from complex datasets (Abideen et al. 2021; Burgos and Ivanov 2021a; Erikstad 2017). Additionally, handling various data formats and reconciling syntactical and semantic discrepancies is essential for accurate analysis and decision-making (Li et al. 2023).

An example of a digital twin in the context of a supply chain is a digital twin of shipments. Finnair Cargo, an air freight carrier, includes the contents of a package or container in its digital twin (Finnair Cargo 2020). If an item is shipped, a digital twin has already been created with data such as its geometry. Alternatively, the item data can be generated when preparing the shipment using 3D scanning. The combination of product and packaging data could help companies improve efficiency by automating packaging selection and container packaging strategies to optimise usage and product protection (Finnair Cargo 2020).

Dachser's warehouse digital twin, on the other hand, presents a virtual mirror of the company's physical warehouse operations. This system utilises real-time data to simulate and optimise various warehouse processes. The implementation of this technology in warehouse management highlights its capacity for detailed monitoring and efficient management of logistics operations. It exemplifies how digital twins can be leveraged to improve operational efficiency and inform strategic decision-making in supply chain systems (Dachser 2023). These examples illustrate the diverse utility of digital twin technology in supply chain management.

Related surveys on SCDT

Research that focuses specifically on supply chain digital twins is scarce. Srai et al. (2019) provide an overview by looking at three industry cases: pharmaceutical, organic food, and precision agriculture. They derive possible attributes from those cases consisting of ambitions, scope, boundaries,

and infrastructure. The attribute scope is further divided into assets, unit ops, network configuration, and multi-echelon inventory and service modelling. The attribute infrastructure is divided into platform technology, supply chain mapping tools, and digital data acquisition management. In addition, they identify opportunities and challenges for the three industry cases. Opportunities include improved service quality, reliable authentication, digital platforms, and resource efficiency. Challenges include end-to-end visibility, the sensitivity of information, lack of agreed infrastructure, product-technology complexity, and technology accessibility and skills (Srai et al. 2019). The authors derive possible attributes of a supply chain digital twin from three industry cases. We include those industry cases in our analysis below. Further review papers focused on operations and supply chain digital twins (Bhandal et al. 2022) and enablers and barriers for digital twins in the process industry (Perno, Hvam, and Haug 2022).

The literature focusing only on the manufacturing part of the supply chain for digital twins provides a larger variety of papers. Those papers focus on specific shop-floor digital twins. Tao and Zhang (2017) provide in their paper 'Digital Twin Shop-Floor: A New Shop-Floor Paradigm Towards Smart Manufacturing' a new concept for shop-floor digital twins. They identify five key technologies. The first key technology is the interconnection and interaction in the physical shop floor, and the second key technology is the modelling, operation, and verification of the virtual shop floor. Other key technologies are the construction and management of digital twin data, as well as the operation and evolution of digital twins. The fifth key technology is smart production and precious services based on digital twin data. Tao and Zhang (2017) subsequently identified the challenges of maintaining adequate two-way connections between physical and virtual spaces. Another challenge is the need for high-fidelity models on the digital twin side to provide a stable foundation for the physical object regarding variability, uncertainty, and fuzziness. Finally, the seamless integration of the digital twin with respect to the large amount of data is an additional challenge (Tao and Zhang 2017).

Other research focuses on production and cyber-physical systems (Negri, Fumagalli, and Macchi 2017; Park, Son, and Noh 2020; Uhlemann, Schock et al. 2017; Zhuang, Liu, and Xiong 2018). Uhlemann, Lehmann, and Steinhilper (2017a) develop a learning factory-based concept for digital twins, demonstrating the potential of real-time data acquisition in production systems. Ivanov et al. (2019) focus on the ripple effect, resilience, and disruption risks by data-driven optimisation, simulation, and visibility in the context of supply chain digital twins. In our paper, we address the research gap for supply chain digital twins. Building on the aforementioned literature, we analyse applications and use cases from industry and academia and conduct expert interviews to develop a framework.

Methodology

Since the goal of our research is to develop an artifact that can be used to solve a problem that is relevant for both science and practice, we applied design science as our methodical approach (Gregor and Hevner 2013; Hevner et al. 2004; Peffers et al. 2007; Winter 2008). In this case, the artifact is a framework for supply chain digital twins. Each research step is illustrated in Figure 2.

Problem motivation

We searched various papers related to the supply chain digital twin to find the current state of research as well as research gaps. The related surveys on supply chain digital twins mentioned in the background chapter describe the content. Due to the minimal amount of research in the supply chain digital twin field, a lack of a holistic overview of supply chain digital twins with different elements has been shown to be helpful in implementation and consideration. Through the existing literature and in casual conversations with experts at conferences, the objective of the framework became clear. The digital twin and its corresponding technologies are in an early phase of their

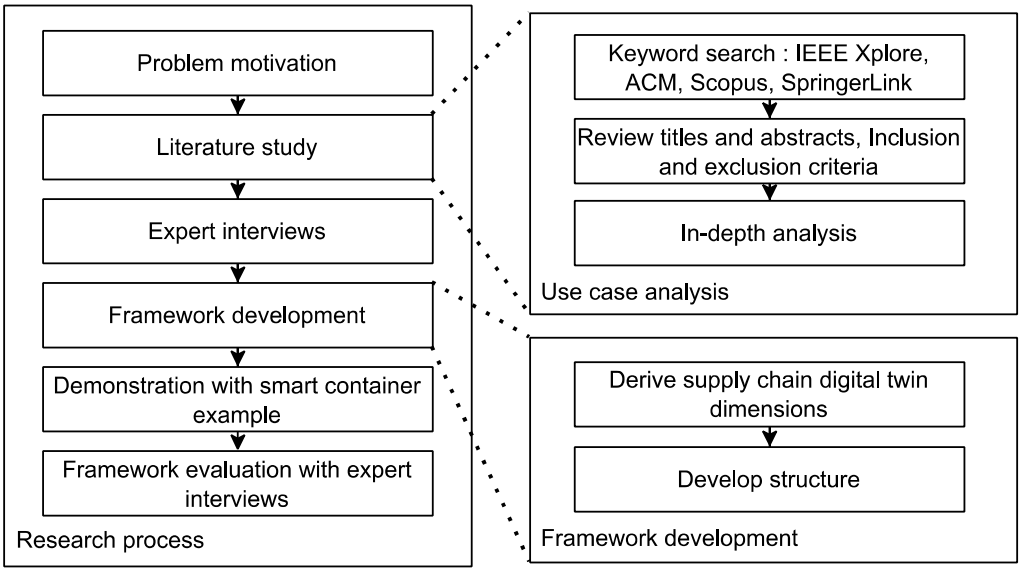

Figure 2. Research process.

development. The challenges in developing and deploying a digital twin are manifold (Tuegel et al. 2011). The digitalisation of the supply chain consists of many challenges (Kache and Seuring 2017). There is a research gap for the specifics of digital twin concepts for different products and industries and a lack of tools to evaluate supply chain digital twin implementations (Holler, Uebernickel, and Brenner 2016). There are challenges at the decision-making level when digital twins are introduced in the supply chain context: What are the essential components of supply chain digital twins? How can the components be structured and how do they relate to each other?

Literature study

To answer the question on the state of the art of supply chain digital twin applications, a literature study was conducted over the last ten years. First, we separated the technical approaches from the conceptual approaches and analysed the various applications and use cases. The results can be found in the section Use Case Analysis based on a Literature Study. A further objective of this study was to specify relevant supply chain dimensions and develop a classification scheme based on existing supply chain digital twin research (Ivanov et al. 2019; Ivanov and Dolgui 2019; Srai et al. 2019).

We conducted our search in different steps to identify relevant peer-reviewed articles on supply chain digital twins, see Figure 3. In the first step, we conducted the snow-balling technique on the

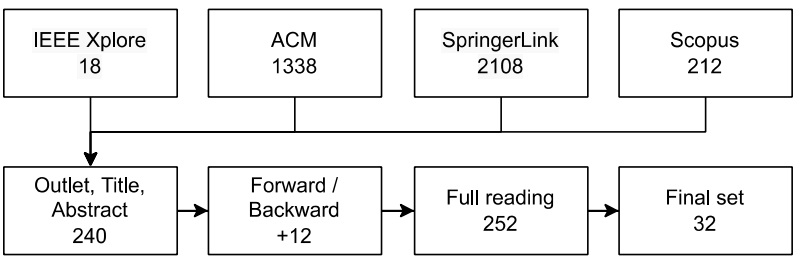

Figure 3. Literature selection process.

existing literature reviews on the subject. In the second step, we searched the following databases: IEEE Xplore, ACM, Scopus, and SpringerLink. We used combinations of the keywords: digital twin + supply chain. We extended the keywords by synonyms used for digital twins in the industry: smart logistics and logistics intelligence. Both terms refer to technical concepts similar to those of supply chain digital twins, and there is a large overlap between the different terms (Uckelmann 2008; Y. Wang et al. 2014). In addition, we looked at similar concepts, such as supply chain control tower and supply chain analytics. In the search for use cases from the industry, we looked at university repositories, industry white papers, and short technical descriptions on project landing pages. Many industry applications focus on specific digital twin features. We decided to include these articles, as we consider that the information provided is a valuable use case.

In total, we obtained 252 papers. To evaluate the search results, we defined the inclusion and exclusion criteria for considering a paper or industry use case for our study. These criteria provide a broad foundation to select state-of-the-art papers and use cases. For a paper to be included in our result set, the title and abstract of the paper or the description of the use case must indicate that the work considers the application of digital twin technologies on some type of supply chain. Papers that were not accessible or not available as full-paper documents were excluded from the result set. These criteria provide a broad foundation to select state-of-the-art papers and use cases. They comprise the following inclusion and exclusion criteria: The paper should be published after 2010. From the title and abstract of the paper or the description of the use case, it must be clear that the work considers the application of digital twin technologies on some type of supply chain. Non-accessible paper or non-available full-paper document. The paper length is less than 4 pages.

There are possible limitations to our research methodology. However, we believe that our approach has brought together the critical mass of applications and use cases on digital twins in supply chains that exist both in academia and industry, allowing us to carry out an in-depth analysis of the topic.

Expert interviews

Expert interviews were conducted to gather qualitative requirements and further define the artifact. We conducted five interviews in April and May 2021. The experts were known to us from presentations at conferences and workshops on digital twins in supply chains, and we were able to approach and interview them in person. The experts are from different sectors, such as logistics service providers, IT, and consulting. The interviews were conducted online via Zoom or Microsoft Teams and lasted on average of 20 min. We used an interview guide to conduct the semi-structured interviews. We asked the participants about their definition of supply chain digital twins and the difference from other technologies and concepts. In addition, we asked specific application questions about how the digital twin is used in their company, how it is integrated into existing processes, and what applications are used for the digital twin. Furthermore, questions regarding the goal and tasks of the digital twin in a company were named, and the differences regarding the maturity level of digital twins were questioned. The same interviewees were confronted with the evaluation of our developed framework in the second part of the interview. Each interview was transcribed and coded, and then the results of the expert interviews were discussed by the author team.

Framework development

We iteratively designed and improved our artifact based on the use cases assessed in the literature study and the output of expert interviews. First, based on the use cases analysed in the literature study, relevant dimensions (Scope, Actor, Asset, Flow Reference Object, Performance Measurement, and Supply Chain Process) of supply chain digital twins were grouped and derived. A literature analysis was conducted to investigate what previous frameworks exist in the field of digital

twins and what elements they contain. In addition, the elements associated with the dimensions were derived from the use cases and existing artifacts. Based on these findings, the dimensions were further improved and extended with the help of expert interviews. Afterward, a corresponding framework was developed. In the course of the project, the framework was further adapted. In addition, a layered view was chosen for the framework to better structure the dimensions. Finally, cross-layer elements were identified based on feedback from the expert interviews.

Framework evaluation with expert interviews

To evaluate the artifact, we follow the framework for evaluation in design science research (Venable, Pries-Heje, and Baskerville 2016). The evaluation is carried out with participation from companies. There, using an illustrative example the artifact is evaluated against different criteria (completeness, level of detail, generality, simplicity, correctness) with expert interviews. Interviews are a standard approach in the evaluation of frameworks in the context of design science (Labazova 2019; McNaughton, Ray, and Lewis 2010; Poeppelbuss and Lubarski 2019; Schilling and Winter 2019).

Use case analysis based on a literature study

We classified the different elements of the supply chain based on different dimensions and corresponding characteristics in our previous work, as outlined in Table 1 (Freese and Ludwig 2021).

Qualitative requirements analysis

In addition to the literature research, the requirements for the framework were determined using expert interviews. The semi-structured interviews were conducted with five experts from four different companies working in the supply chain sector. The focus of the experts was based on their area of application of the digital twin in production, warehouses, or logistics chains.

The expert interviews enabled an in-depth analysis of the supply chain digital twin dimensions. The experts were asked about the specifics of a digital twin in the supply chain as well as the goals and tasks of digital twins in the supply chain and differentiation from other similar terms. Subsequently, questions on the implementation and use of digital twins were discussed. The integration into the existing process landscape was discussed in detail by the experts. Furthermore, concrete applications were discussed. The experts were also asked about the different maturity levels of a digital twin. We follow the approach to describe the expert interviews as Miehle et al. (2020).

We interviewed the team lead for the Internet of Things division of a railway company as an expert **(Interview [I1])**. Concrete examples in his field of work include the modelling of network and track infrastructure in a digital twin, as well as building infrastructure modelling. This corresponds to one of the use cases of digital twins we analysed in our paper.

Table 1. Classification scheme of the supply chain in relation to digital twins.

Dimension	Characteristics
Coverage (Barratt and Barratt 2011)	Internal Supply Chain, External Supply Chain
Actor (Giannoccaro and Pontrandolfo 2004)	Shippers, Freight Forwarders, Manufacturers
Asset (Mason et al. 2003)	Assets at network nodes (i.e. warehouses, plants, handling equipment), Assets at network links (i.e. vehicles, trucks, trains, containers)
Flow reference object (Pfohl and Gomm 2009)	Goods (i.e. stock), Information (i.e. status data, stock levels), Finances (i.e. credits, invoices), Liabilities (i.e. bill of lading)
Performance measurement (Chan 2003)	Costs, Quality, Resource Utilisation, Flexibility, Visibility, Trust, Innovativeness
Supply chain process (Mentzer et al. 2001)	Demand Planning, Procurement Process, Production Planning, Production, Warehouse and Distribution

Another interview **(Interview [I2])** was conducted with an executive of a logistics solution provider that digitalises logistics processes using software. Specialisation on shop-floor digital twins as an aspect of production in supply chain digital twins was also confirmed in the literature. The external supply chain enables a far-reaching visualisation of the supply chain conditions and makes it possible to identify problems at an early stage.

Furthermore, in interview number three **(Interview [I3])**, we interviewed a consultant at a consultancy firm for digitalisation and innovation in supply chain. Another important aspect is horizontal integration so that not only is the company represented in the digital twin but also upstream and downstream areas such as suppliers and customers. As another example of a digital twin, he refers to an example from production.

The fourth interview **(Interview [I4])** was also conducted with a consultant from a consulting company for digitalisation and innovation in supply chain. According to his statement, the challenge for the digital twin is primarily the reconciliation between the real and the digital world. It must be identified who the beneficiary of the digital twin is and what they can do with it.

Finally, we interviewed the head of the data science department at a logistics and freight forwarding service provider **(Interview [I5])**. His main focus for the digital twin is primarily on warehouse management as part of the supply chain. In this way, the locations of the various items can be displayed in the digital twin at any given moment, and key performance indicators can be derived. The reference objects of the supply chain digital twin in this interview are primarily the various parts of the warehouse, such as racks and shelves. In addition, the inventory is needed to replenish them.

Resulting framework

To develop the framework, we use both the elements from the literature study with the corresponding use cases from research and industry and the results of the qualitative requirements analysis based on expert interviews. Based on the derived supply chain digital twin dimensions, a layered model provides the foundation structure of our framework. In addition, we have consulted various existing frameworks in the design decision. For example, the framework from Liu et al. (2019) looks at the digital twin in healthcare and includes a layered structure. Individual dimensions were derived from existing research papers (Enders and Hoßbach 2019; X. Wang et al. 2021). The proposed framework is shown in Figure 4.

The basic structure of the framework is as follows. The left side of the framework contains the various layers with their associated dimensions and elements. On the right side, cross-layer elements are listed that are independent of the other elements and span and influence the layers.

Data layer

The base layer is the data layer with the dimension's origin and quality. The data provide the foundation for the supply chain digital twin framework, and these data can come from various origins, such as production, users, or service (Qi et al. 2021). Production data include, for example, data from machines of all types, from milling machines to forklifts. User data include data from the various actors within the supply chain, such as location data, movement data, and personal process data. Service data provide information about the status of machines and devices needed for service, such as predictive maintenance [I5]. Non-company-specific data include data from the environment, such as weather data or traffic data, as well as data about port and container availability. The quality dimension describes the various properties of the generated data. Depending on the specific digital twin and its application and objective, data of different accuracy, completeness, consistency timeliness, validity and uniqueness are required. For time-critical applications such as decisions by digital twins in the allocation of production resources, the data must be made available in real time (Fuller et al. 2020), [I5].

Figure 4. Supply chain digital twin framework.

Technology and infrastructure layer

The next layer is the technology and infrastructure layer. It contains various technologies for capturing, processing, and displaying the data. It is divided into these three units accordingly. The acquisition technologies include various sensors, such as brightness sensors, shock sensors, and temperature sensors (Fuller et al. 2020), [I1, I5]. Actuators enable the digital twin to control the physical system and react to changes detected by sensors (Rocha et al. 2022). In addition, this element includes various Internet of Things applications as well as RFID and GPS technologies (M. Liu et al. 2021). Additionally, the data acquisition can be done manually via an app using a computer, tablet, or smartphone. The data obtained are processed using various technologies and methods. For example, based on historical data, machine learning algorithms can be used to predict future outcomes. High-performance computers and in-memory databases are used to process a large amount of real-time data [I5]. Blockchain technologies enable decentralised data processing with immutable data storage and processing through smart contracts (Qi et al. 2021). In addition, technologies such as 5G and cloud computing play an important role. The subsequent presentation of the data takes place, for instance, in conventional dashboards or via websites and applications. Corresponding application programming interfaces are used to connect to other information systems. Virtual reality glasses are used to display and develop 3-D models (Perno, Hvam, and Haug 2022).

Application layer

The application layer comprises the dimensions of reference objects, purpose, and performance measurement. The reference object defines the physical objects that are mapped in the digital twin. Thus, digital twins of individual assets such as production machines or high-bay racks can be mapped. A more extensive utilisation of digital twins refers to an entire system in which various assets are interconnected and interact in a value-adding manner via a process-related workflow. These systems can, for example, be production lines in which both production machines and production employees are involved. In the example of the warehouse, a complete warehouse with various high bay racks as well as employees and processes would be mapped in the digital twin. The system-of-systems approach extends the digital twin in a hierarchical way, where individual digital twins interact with each other to form a complex system (Dietz and Pernul 2020; Zhang et al. 2021).

For example, different warehouses are connected via different transport vehicles, such as trucks, planes, or ships [I5]. The purpose of the dimension defines the intention behind the application of the digital twin (Enders and Hoßbach 2019). Simulations can be realised with the digital twin. This includes, for instance, what-if analyses in which hypothetical assumptions and their effects on complex systems can be examined (M. Liu et al. 2021), [I2]. Monitoring is about the observation of different states of the supply chain. This enables high transparency and better decision-making. Control is about detecting deviations from predefined key figures or processes in real time, if feasible, to be able to intervene in a steering function. The forecasting dimension refers to future predictions, including the time of arrival forecasts. The digital twin enables reliable forecasts to be made using the data provided [I1]. The dimension performance measurement describes the measurement possibilities that can be assessed and improved through the use of a digital twin (Chan 2003). On the one hand, costs and quality can be monitored through the digital twin. On the other hand, the use of resources can be measured by assessing, for example, if resources are being used correctly or whether there are idle times or occurring issues (Fuller et al. 2020). Another goal of the digital twin can be the flexibility it offers. New products can be tested quickly and changed in a flexible way. Visibility is also an important aspect. Through the high number of generated data and their depiction in an understandable format, the visibility over the supply chain can be increased (M. Liu et al. 2021). Other aspects are trust, which can be generated by the digital twin. In connection with transparency, the end customer has the opportunity to view the origin and production data of products through supply chain digital twins. Furthermore, the ability to innovate is also an important aspect. By using modern technologies that are connected to the digital twin, the innovation power of the company is boosted, which can also be an important objective.

Workflow logic layer

Another layer of the framework is the workflow logic. This contains the processes in which the digital twin is integrated. In the case of the supply chain digital twin, these are the supply chain processes. Supply chain processes can be roughly divided into demand planning, procurement, production planning, production, warehousing and distribution, and customs clearance (Mentzer et al. 2001). Depending on the use of the digital twin in the supply chain, the affected processes are mapped and integrated into the digital twin [I3, I4, I5].

Structural model layer

Another layer is the structural model. This layer contains specific elements for the supply chain digital twin. On the one hand, the coverage dimension describes the extent to which the supply chain is covered by the digital twin (Barratt and Barratt 2011). A distinction must be made here as to whether only the internal supply chain is mapped by the digital twin or extends to the external supply chain [I2]. For the external supply chain, it must be decided how in-depth the digital twin should go. The digital twin can contain only direct supplier companies or go far-reaching depending on the tier up to the n tier to the original production. Looking in the other direction, the digital twin then extends to the end customer [I3]. The actor dimension describes the involved stakeholders within the digital twin. In the supply chain digital twin, these include shippers, freight forwarders, and manufacturers (Giannoccaro and Pontrandolfo 2004), [I5]. The asset dimension describes the individual entities that can be mapped in the supply chain digital twin (Mason et al. 2003). Using the network concept with different systems, the network nodes are represented by warehouses, plants, and handling equipment [I5]. Network links include trucks, ships, airplanes, and trains [I1]. Another dimension in the structure model layer is the flow reference objects. These are the objects flowing in the supply chain network (Pfohl and Gomm 2009). These include the goods that are produced and transported within the supply chain and the information about the goods and processes [I5]. Another reference object is finance, which represents the monetary aspect

of the supply chain and is exchanged in return for these goods. An additional reference object is liabilities, which describe the contractual conditions within the supply chain [I2]. Table 2 provides an overview of the different dimensions and their description.

Cross-layer elements

In addition to these layer-based dimensions, there are other elements that are independent of them. One of the elements is the maturity level of the digital twin. This defines how advanced the digital twin is (Madni, Madni, and Lucero 2019). At a low maturity level of description, only simple descriptive activities can be performed. At the diagnostic maturity level, diagnoses can be made based on historical data. The next maturity level predictive enables the forecasting of future problems. The highest maturity level prescriptive enables the digital twin to anticipate and intervene and automatically influence the physical object. At this high maturity level, actuators are needed in addition to sensors [I2, I5]. Another element is level, which describes the scale at which the digital twin is used. At the lowest level, only individual assets are mapped in the digital twin, such as a single production machine. At the next level, the production line, more objects are mapped in the digital twin. An even higher level is the location where not only a production line, but an entire production site is mapped as a digital twin [I2, I3]. The connection of several locations to a network represents the highest level of the digital twin. Complex digital twins act together in a network-like structure. The creation time element describes when the digital twin is created (Enders and Hoßbach 2019). There are two possibilities. Either the digital twin is created before the physical object, i.e. already in the planning phase of the physical object. Alternatively, the digital twin is created after the physical object. This means that a physical object already exists, such as a high rack in the warehouse, which is then mapped as a digital twin using the corresponding sensors [I5]. The connection element describes the connectivity of the digital twin to its physical object (Enders and Hoßbach 2019). In the simplest case, there is no direct connection between the physical object and the digital twin. The digital twin is created independently from the physical object. Another possibility is a one-directional connection from the physical object to the digital twin, also known as digital shadow (Kritzinger et al. 2018). In this case, the physical object delivers data to the digital twin via the sensors, but the other direction of the connection from the digital twin to

Table 2. Framework dimensions.

Dimension	Description	Reference
Coverage	The different reach of the digital twin, if the digital twin is just for the internal supply chain of a company or if it is extended to external companies' supply chain.	[I2, I3], Barratt and Barratt (2011)
Actor	Decision-makers pursuing different objectives.	[I5], Giannoccaro and Pontrandolfo (2004)
Asset	Refers to any kind of asset that interacts between the various supply chain processes.	[I1, I5], Mason et al. (2003)
Flow reference object	Describes objects like goods, information, and financial values. Flow reference objects move along the process stages of the supply chain.	[I2, I5], Pfohl and Gomm (2009)
SC process	Describes the documented procedures in the course of the supply chain.	[I3, I4, I5], Mentzer et al. (2001)
Reference object	Describes which component a digital twin refers to.	[I5]
Purpose	The purpose of digital twins can be divided into three main characteristics: simulation, monitoring, and control.	[I1, I2], Enders and Hoßbach (2019)
Performance measurement	Describes the criteria to evaluate supply chain performance. The most common performance measurements are cost and quality. Other performance measures are resource utilisation, flexibility, visibility, trust and innovativeness.	Chan (2003)
Origin	Describes the source of the data, which is then aggregated in the digital twin.	[I5], Qi et al. (2021)
Quality	Describes the level of quality of the data. Depending on the requirements, different data quality is needed.	[I5], Fuller et al. (2020)

the physical object does not exist [I1]. However, in a bi-directional connection, communication is possible via sensors and actuators from the physical object to the digital twin, and vice versa.

Demonstration of the framework

To demonstrate the use of the framework, we explain how it can be used. In our example a smart container should be the real object to be presented by the digital twin. The following describes how a supply chain planner would use the framework in order to create a digital twin. The smart container is defined as a standard shipping container, as it is usually used in maritime logistics, with additional features. The container is equipped with a variety of sensors that provide information about the condition of the container. To integrate and use this smart container in an existing supply chain, the supply chain digital twin framework can be used to learn about the various needs, capabilities and opportunities and to provide decision and implementation support.

A logistics company decides to map the containers to be shipped with the help of a digital twin in order to increase the visibility and traceability of the supply chain. Now the supply chain planner has to decide what is important for the digital twin of the container, Figure 5.

1. Starting with the base layer *data*, it is necessary to consider which data is relevant based on the *origin* dimension. In the container example, service data that provides the customer with additional information about expected arrival times is relevant. In addition, the data *quality* is delivered as timely as possible in order to provide the customer with up-to-date information about his delivery.
2. The next layer *technology and infrastructure* helps to decide how the required data is collected, processed and displayed. In the smart container example, GPS sensors, placed outside on the container, are used to provide feedback on the location. For data *processing*, a corresponding database must be provided, as well as machine learning algorithms for making predictions of the expected arrival. The *presentation* of the information is integrated on the existing company website as an additional feature.
3. In the *application* layer, on the one side, the *reference object* is to be determined. In this case it is an asset – a container. The main *purpose* of the application is to monitor and forecast the delivery of a container. The main objective to measure the *performance* of the digital twin here is to focus on the visibility of the container in the course of the supply chain.

Figure 5. Framework demonstration.

4. The *workflow logic* layer defines the processes into which the container is integrated. The container is integrated into the shipping process and the digital twin is complementarily embedded into the existing process landscape.
5. Finally, the *structural model* layer describes the *coverage* of the digital twin. The smart container is handled in the internal and external supply chain of the logistics company. Therefore, the partner companies must be involved in the digital twin to facilitate the external supply chain coverage. The actors involved with the smart container are the shippers and freight forwarders. Furthermore, the different *assets* are identified. Here the assets contain only the containers. The *flow reference objects* in the structural model are the containers and their corresponding information.
6. The framework includes cross layer elements. The digital twin of the container predicts the expected time of arrival and therefore corresponds to the predictive *maturity*. The digital twin is at asset level and is created after the corresponding physical object. The *connection* is one-directional from the container to the digital twin.

Evaluation against criteria

Following several rounds of internal adjustments to the framework in terms of structure and content, the framework was ready for evaluation. We chose expert interviews as the evaluation method because this evaluation strategy allows us to evaluate how well the framework is received and understood by supply chain experts (Sonnenberg and Brocke 2012; Venable, Pries-Heje, and Baskerville 2016). Experts are drawn from different areas of the supply chain, allowing for a broad knowledge of how digital twins can be used in the supply chain. The hierarchy of evaluation criteria by Prat, Comyn-Wattiau, and Akoka (2015) was used as a reference source to define the aspects to be covered by the framework. Considering the evaluation criteria already used from other literature (Blaschke et al. 2019; Schilling and Winter 2019), we decided to evaluate the criteria completeness, level of detail, generality, simplicity, and correctness (see Table 3).

The semi-structured interviews were conducted with five experts. First, they were introduced to the framework and then asked criteria-based questions. The individual interviews lasted an average of 38 min. All interview partners liked the idea of providing a framework for digital twins in the supply chain. Regarding the criterion of completeness, most interview partners confirmed that all relevant elements were mentioned. Responding to feedback highlighting the importance of the time dimension, we integrated 'timeliness' into the quality dimension of our framework. This addresses the varying time requirements between production-focused digital twins, which need real-time data, and those for supply chain planning with less time-sensitive data needs. Regarding the criteria correctness and simplicity, the requirements for the framework were also met. With regard to the level of detail, one interviewee mentioned further options for distinguishing the depth of the digital twin and different tiers. For the generality criterion, the customs aspect was mentioned as an important application area of the supply chain digital twin. Based on the aforementioned feedback, the final framework was established.

Implications and guidelines

In this section, implications for theory and practice are listed, and guidelines based on the framework are developed. Various findings and implications for practice emerge from the results of this work.

Supply chain digital twins may cover both internal and external processes.

The digital twin for the supply chain and logistics must map the entire necessary coverage and internal and external supply chain processes, including potential specific logistics services such as customs clearance [I2]. The effects of disruptions can thus be mirrored and recognised in other areas of the supply chain (Burgos and Ivanov 2021b). This is in contrast to the digital

Table 3. Evaluation with expert interviews.

Evaluation criteria and exemplary questions	Illustrative answers
Completeness: Are all relevant elements of Supply Chain Digital Twins included in the framework?	[I4]: 'First, I have to say, hats off! That's quite something. It goes in a similar direction. … I think it's very good. I like that very much'. [I2]: 'So for me, what is still, what I am also missing a bit, is a time dimension'. [I5]: 'So this list is very extensive, and I like that'.
Level of Detail: Should a dimension in the framework be more detailed?	[I4]: 'So for now, that's, that's speaking enough. … So from my point of view, it fits pretty well here'. [I2]: 'Well, it depends now … So I could imagine, for example, with Scope, you now write internal and external supply chain. So if I think of an automotive company, yes, they control the external supply chain in procurement down to the Tier 5 Tier 6 level'.
Generality: Can the framework be applied to all supply chain digital twins?	[I4]: 'So essentially, you can also apply this to any digital twin in the generics'. [I2]: 'So the first thing that would come to my mind is another one. That is now, or are actually, two topics, but they go to the same thing. So, what else? Another interesting thing is the issue of customs'. [I5]: ' … probably you can use it'.
Simplicity: Do you have any suggestions for further simplification/improvement of the framework?	[I2]: 'So from my point of view, I think that the things with the framework presented are already appropriate and necessary to describe this'. [I5]: 'It's very extensive, and you covered many aspects, so I think from that perspective is good enough. … Are all of the questions and the aspects covered here? Probably, yes'
Correctness: In your opinion, are the different dimensions of the framework correct?	[I1]: 'The structure you present is valid, yes' [I4]: 'That also fits at first sight'. [I5]: 'I also agree, monitoring, agree, forecasting, agree, control. … Yeah, the levels of purpose, how much you intervene in the system. How much action you make on the system. Yeah, OK, performance, cost, quality'.

twins in the production or construction sector. Many dimensions were not yet known and were not yet mapped in a framework. The supply chain digital twin maps all processes along the supply chain. For instance, a digital twin in a global food distribution company could track and manage everything from the sourcing of agricultural products to their delivery at retail outlets. This twin would integrate customs clearance processes for efficient international trade compliance. In scenarios like weather disruptions or geopolitical issues, the digital twin's extensive coverage allows for swift strategizing of alternative solutions.

Digital twins entail not only sensor technology and data acquisition but also applications.

The digital twin covers the data acquisition and mirroring of the physical object of the real-time events and applications such as forecasting, simulation, or error management. In contrast to current work, for instance, Uhlemann, Schock et al. (2017) focused on the data acquisition side of a digital twin. It is also necessary to have the applications that use that data in mind. Digital twins entail not only sensor technology and data acquisition but also applications (Fuller et al. 2020; M. Liu et al. 2021). These depend on the purpose, a holistic view of the process, purpose within the process, and systems and data. Collecting data from the digital twin and mirroring the physical object is not enough. The data must be used in a goal-oriented manner for the needs of the company (Qi et al. 2021), [I3]. Based on this, applications such as error management, simulation, and forecasting are provided by the digital twin.

In logistics, a digital twin can be used for optimising delivery routes and predicting maintenance needs. By analysing real-time data from sensors on delivery vehicles, the digital twin can identify maintenance requirements before they become critical issues, minimising downtime. It can also

simulate various routing options to find the most efficient paths, considering dynamic variables like traffic and weather.

In a manufacturing context, a digital twin might be used for enhancing product design and testing. By simulating manufacturing processes and conditions, it can identify potential design flaws or production bottlenecks, allowing for adjustments in advance of actual production. This leads to a more efficient production process, reducing waste and improving overall product quality.

These instances demonstrate the diverse capabilities of digital twins, highlighting their role not just in data gathering but in applying this data intelligently to improve business processes and strategic decision-making.

A one-directional or bi-directional digital twin can be necessary depending on the maturity level.

Depending on how the digital twin is used, different connection types are necessary. If the digital twin is used to monitor a specific asset such as a truck, for example, a one-directional digital twin is sufficient (Enders and Hoßbach 2019). Here, the tracking data of the truck are reflected by the digital twin without the need to redirect or reroute the traffic. On the other hand, a digital twin is needed for monitoring and controlling a bi-directional connection. If the real object is to be actively intervened in and steered via the digital twin, a bi-directional twin is required. A higher maturity level needs a bi-directional digital twin [I2, I5].

All the required actors must be kept in mind, and their information needs to be able to decide which processes are mapped in the digital twin. Referring back to the case studies analysed in the literature study, an example is ETA prognosis. If the digital twin is of a descriptive maturity level for an ETA prognosis, the time measurements should be reported (Deutsche Post DHL Group 2019). This requires a one-directional connection and certain data points from time sensors. For a digital twin with a prescriptive maturity level, a bi-directional connection is indispensable. Therefore, to administer interventions, it additionally requires a connection from the digital twin to the physical object. In the example of ETA forecasts, the digital twin delivers prescriptive instructions to the physical object, such as route commands, and can thus influence its behaviour.

Depending on the coverage of the supply chain, the actors involved must be selected. This determines which processes the digital twin must support

The coverage of the digital twin forms the basis for selecting the actors involved. In addition, the coverage of the digital twin determines which relevant processes are mapped in the digital twin or which interfaces exist. In the case of a digital twin with low coverage as a subsection of the internal supply chain, the relevant actors are elicited and used to develop the digital twin [I2]. An example would be the introduction of the digital twin in a production line. Here, the coverage of the digital twin is clearly defined to one production line. The actors are the production workers. The relevant processes are the processes defined for this production line [I4]. This example shows that the coverage of the digital twin determines the corresponding actors and processes.

In contrast to digital twins with limited coverage focused on a subsection of the internal supply chain, cross-enterprise digital twins represent a broader scope, encompassing multiple stages of the supply chain across different organisations. This wider coverage significantly influences the selection of involved actors, expanding it to include various external stakeholders. In the automotive industry, a cross-enterprise digital twin extends its functionality from raw material suppliers to end customers. It ensures efficient material deliveries to manufacturers, monitors production processes, and coordinates logistics for vehicle delivery to dealerships. At the dealership level, it aids in inventory management and improves customer service. Post-sale, the digital twin engages with customers by tracking vehicle performance and providing maintenance updates. This holistic approach demonstrates the digital twin's capability to synchronise complex networks across the entire supply chain.

Depending on the processes, the different actors and assets are to be considered

The processes represented in the digital twin determine the actors and assets involved. In the case of a digital twin, integration into the existing process landscapes is an essential part of adoption. This requires identifying the processes impacted by the digital twin and the actors and assets involved in these processes [I3]. If, for example, the delivery process is to be mapped via a digital twin, the actors involved, such as delivery personnel, must be identified and involved in the development. In addition, the relevant assets must be identified, such as the parcels to be handled and the trucks involved in the delivery process. In a warehouse setting, the digital twin would involve warehouse staff, inventory managers, and logistics coordinators as actors, with assets including the warehouse infrastructure and inventory items. This twin monitors inventory levels, tracks goods movement, and optimises operations, providing a comprehensive operational view for improved decision-making and efficiency.

Depending on the purpose of the digital twin, different connections and technologies are necessary

For a supply chain digital twin with the intention of monitoring, a one-directional connection from the physical object to the digital twin is necessary. This one-directional connection is accomplished with appropriate technologies such as sensors (Enders and Hoßbach 2019). For example, delivery trucks and their position and delivery route can be monitored using appropriate sensors. As a technological basis, these require GPS sensors and an internet connection to transmit the data (Qi et al. 2021). Furthermore, with the purpose of control with the use of the digital twin, a bi-directional connection from the physical object to the digital twin and vice versa is necessary. Thus, as in the previous example, sensors are needed to transmit data from the physical object to the digital twin. To control the physical object with the help of the digital twin, corresponding actuators are required (Qi et al. 2021), [I2]. In the example of the delivery truck, this can be done via an app that displays the optimised route. The driver must then manually follow the revised route.

The actuators can also communicate automatically with the physical object by programming interfaces and controlling its behaviour. For example, a smart manufacturing setup requires a bi-directional digital twin. Here, sensors collect data from machinery about operational status and output. The digital twin not only analyses this data but also actively intervenes. If it detects a machine operating suboptimally, it can autonomously adjust the machine's settings or halt it to prevent inefficiency. This level of control is achieved through advanced technologies like programmable logic controllers and AI algorithms, showcasing the digital twin's capacity for real-time decision-making and process optimisation in a manufacturing environment.

Conclusion and outlook

Based on a systematic literature review and expert interviews, we identified and analysed different applications and use cases for supply chain digital twins. We analysed the existing research approaches on the subject of digital twins for supply chains. We developed a conceptual framework with the relevant supply chain dimensions to answer the question of which elements of the supply chain can be represented by digital twins.

Supply chain digital twins are just in their infancy. Although the concept of digital twins has been around for many years, the use of supply chain digital twins is quite new. Only in the last year have researchers and practitioners started to cultivate this topic. In our paper, we provide an overview of these first approaches and categorise them based on relevant supply chain dimensions. Building on previous supply chain digital twin research (Ivanov et al. 2019; Ivanov and Dolgui 2019; Srai et al. 2019), we provide general dimensions for digital twins in the context of the supply chain. In addition, qualitative requirements were gathered through expert interviews and subsequently a

conceptual framework was developed. Afterwards, the framework was evaluated and the relationships between the individual dimensions and their implications were analysed.

To provide practical applications from our research, we propose a set of managerial guidelines for organisations venturing into digital twin initiatives in their supply chains. It is crucial for managers to establish coverage that aligns with the digital twin's operational objectives. They should tailor the technology to suit monitoring or control purposes and ensure seamless integration with current processes, involving all relevant actors and assets. Additionally, fostering collaboration among stakeholders across the supply chain is essential. Managers need to harness the twin for in-depth data analysis and informed decision-making while prioritising employee training and robust cybersecurity measures to maximise the benefits of digital twins in supply chain management. Managers should be aware of potential pitfalls. Insufficient planning and strategy, underestimating data quality and integration challenges, and failing to manage organisational change effectively can all hinder the success of digital twin initiatives. Neglecting stakeholder involvement, lacking skilled personnel, and overlooking scalability and flexibility can also limit the technology's potential benefits.

This paper has limitations, as the literature search was limited to IEEE Xplore, ACM, Scopus, and SpringerLink with defined keywords. Even though many use cases were identified, additional use cases may exist that could not be found with the search engines and keywords used. For example, specific terms that are only used in individual industries, i.e. food tracing in agriculture, may be a reason why use cases were not discovered. Furthermore, use cases may exist that have not yet been published in scientific papers or reported by companies.

Regarding the analysis of the use cases, there is a threat to validity in evaluating the individual use cases. The assignment of the individual use cases to the dimensions was carried out by two independent domain experts; however, there is still the risk of subjective assessments leading to a threat to validity.

The framework offers the foundation for further research needs. The individual dimensions require further detailed consideration. On the one hand, the strongly process-oriented approach of the digital twin in the supply chain emerges as an important further research area. The embedding and integration of digital twins in existing process landscapes pose interesting challenges. Further research can build on these dimensions and identify and analyse design goals, interoperability, and interorganizational integration issues. We call for future work to investigate the different supply chain digital twin dimensions more deeply. In particular, the underrepresented characteristics need further investigation, e.g. how financial aspects can be represented in supply chain digital twins.

The participants of this study did not give written consent for their data to be shared publicly, so due to the sensitive nature of the research supporting data is not available.

Disclosure statement

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